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Tech Solutions Pvt Ltd

INFRASTRUCTURE BUDGET DOCUMENT · PREMIUM vs FREE STACK · 2026

Smart Governance System

Infrastructure Cost Comparison — Premium vs Free Tier

■ PREMIUM STACK

Best-in-class infrastructure

High availability · Auto-scaling

99.9% uptime SLA · Dedicated support

Ideal for Assembly constituency

■ FREE / LOW-COST STACK

Maximum free-tier utilisation

Zero or near-zero monthly cost

Reliable for low-medium volumes

Ideal for Panchayat / Municipality

■■Panchayat

Premium: ■3,200/mo

Free: ■590/mo

Savings shown inside →

■■Municipality

Premium: ■14,800/mo

Free: ■1,280/mo

Savings shown inside →

■■Assembly

Premium: ■42,500/mo

Free: ■9,700/mo

Savings shown inside →

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This document provides a complete, line-item budget for the Smart Governance System under two distinct infrastructure philosophies. Both stacks deliver the **same features** to the end user — the difference is entirely in the underlying hosting, AI, and database services used.

PREMIUM STACK

Enterprise-grade services with SLAs, auto-scaling, dedicated support, and the best available AI models. Pay more — worry less. Best suited when reliability and performance are non-negotiable.

AWS / GCP hosted services
OpenAI GPT-4o (best AI accuracy)
Supabase Pro / AWS RDS
Twilio (reliable SMS + voice)
Render Pro / AWS EC2
Cloudinary Pro + backups

FREE / LOW-COST STACK

Maximises every available free tier. Delivers identical features at a fraction of the cost. Some free tiers have usage limits and cold-start delays. Best suited for Panchayat and Municipality scales.

Railway.app free tier (backend)
Gemini 1.5 Flash (free, 1M tokens/day)
Supabase free (500MB DB)
MSG91 (cheapest SMS at 0.16)
Vercel free (unlimited frontend)
Cloudinary free (25GB photos)

§2. COMPONENT-LEVEL COMPARISON — EVERY SERVICE

The table below lists every infrastructure component, the service used in each stack, and the monthly cost at each tier. Costs marked **0** use the free tier.

Component	Premium Service	Free Service	Prem/mo	Free/mo	Who Wins
Backend Hosting	Render Pro (4GB RAM, auto-scale)	Railway.app free (512MB)	4,150	0	Free (low vol)
Database	Supabase Pro (8GB RAM, 100GB, backups)	Supabase Free (500MB, 50K rows)	2,075	0	Free (<50K rows)
AI Classification	OpenAI GPT-4o (99%+ accuracy)	Gemini 1.5 Flash (93% acc, free 1M/day)	5,810	0	Free (93% = fine)
Frontend Hosting	Vercel Pro (SSR, team, analytics)	Vercel Free (unlimited deploys)	1,660	0	Free (same UI)
SMS / OTP	Twilio SMS (0.70/SMS, high delivery)	MSG91 (0.16/SMS, DLT registered)	4,375	1,000	Free (same delivery)
WhatsApp API	Twilio WhatsApp (managed, enterprise SLA)	Meta Cloud API (official, free 1K/mo)	2,490	175	Free
File Storage	Cloudinary Pro (100GB, CDN, transforms)	Cloudinary Free (25GB, 25K transforms)	2,490	0	Free (<25GB)
Session / Cache	Redis Enterprise (Upstash Pro)	Upstash Redis Free (10K cmds/day)	830	0	Free (low vol)
Voice-to-Text	Google Cloud Speech (best accuracy)	Google Cloud Speech (same API)	1,245	415	Same API

Maps / Heatmap	Google Maps Platform (no limits)	Google Maps JS (28K free loads/mo)	2,075	0	Free (<28K)
Monitoring	Sentry Pro + Datadog	Sentry Free + UptimeRobot Free	1,660	0	Free
Email Alerts	Resend Pro (50K emails/mo)	Resend Free (3K emails/mo)	1,245	0	Free (<3K)
CI/CD	GitHub Actions Pro	GitHub Actions Free (2K min/mo)	830	0	Free
Domain	.in domain (GoDaddy premium)	.in domain (cheapest registrar)	125	60	Depends
SSL Certificate	Paid wildcard SSL	Let's Encrypt (free, auto-renew)	830	0	Free
Emergency Calls	Twilio Voice (2/min, high quality)	Twilio Voice (same — Phase 2 only)	830	415	Same
MONTHLY TOTAL (Panchayat)			**3,200**	**590**	**Free saves 2,610**
MONTHLY TOTAL (Municipality)			**14,800**	**1,280**	**Free saves 13,520**

— TIER 3: ASSEMBLY CONSTITUENCY

Full MLA constituency · ~2,00,000 citizens · 30 wards · 800–1,500 complaints/month. Premium stack — maximum performance, zero compromise.

ASSEMBLY — PREMIUM STACK

One-Time Development Cost:

Development Item	Cost (INR)	Notes / Stack Choice
WhatsApp Bot (All Features + Voice)	38,000	GPT-4o, full voice-to-text, emergency mode
Backend API (Enterprise Grade)	30,000	AWS EC2 / Render Pro, clustering, Bull MQ
MLA Dashboard (Full + Predictive AI)	40,000	All analytics + Prophet ML predictions
Ward Dashboard (Complete)	10,000	All features — proof upload, ratings, map
GPT-4o AI Full Pipeline	35,000	Full NLP, telugu fine-tune, duplicate detection
Broadcast + Escalation + Emergency	25,000	Twilio Voice calls, WhatsApp broadcast engine
Transparency Portal + MLA Website	included	Full Next.js + Cloudinary Pro CDN
Security Audit + Pen Test	internal	OWASP ZAP, penetration testing, VAPT report
Load Test (1K concurrent users)	5,000	k6 + AWS CloudWatch analysis
Project Management + Training	2000	Dedicated PM, 4-day training, video guide
Contingency (10%)	30,000	Assembly-scale buffer
TOTAL ONE-TIME COST	**2,50,000**	

Monthly Recurring (Operational) Cost:

Service	Provider	Monthly Cost	Why This Choice
Backend	Render Pro 2× instances	8,300	High-availability, auto-scaling
Database	Supabase Pro + read replica	4,150	Read replica for dashboard queries
AI	OpenAI GPT-4o	8,300	~5,000 requests/month at assembly scale
WhatsApp API	Twilio WhatsApp	3,320	Managed, enterprise delivery guarantee
SMS / OTP	Twilio SMS	8,300	0.70/SMS × ~12,000/month
Voice (Emergency)	Twilio Voice	1,660	~830 emergency calls/month
Maps	Google Maps Platform	2,490	Assembly exceeds free 28K quota

File Storage	Cloudinary Pro (100GB)	2,490	High-volume photo storage + CDN
Monitoring	Sentry Pro + Datadog	2,490	Full APM, error tracking, dashboards
Cache	Upstash Redis Pro	1,245	High-throughput, 1,500+ daily active users
Email + Extras	Resend Pro + domain	1,245	High-volume admin reports + .in domain
Miscellaneous	Buffer	830	API overages, unexpected scaling
TOTAL MONTHLY		44,820/mo	

ASSEMBLY PREMIUM STACK: One-Time Dev = 3,20,000 | Monthly = 44,820/mo | Year 1 Total = ~8,57,840 | Year 2 onwards = ~5,37,840/year

Full MLA constituency · ~2,00,000 citizens · 30 wards · 800–1,500 complaints/month. Free stack at assembly scale — some paid services unavoidable but still 78% cheaper.

ASSEMBLY — FREE STACK

One-Time Development Cost:

Development Item	Cost (INR)	Notes / Stack Choice
WhatsApp Bot (All Features)	25,000	Extend municipality bot — no rebuild
Backend API (Production)	18,000	Render Standard, existing codebase
MLA Dashboard (Full + Predictive)	22,000	Extend municipality dashboard + Prophet AI
Ward Dashboard	12,000	Minor assembly-specific additions
Gemini Flash + Prophet AI	20,000	Free Gemini + open-source Prophet ML
Broadcast + Emergency System	12,000	Meta broadcast + Twilio only for emergencies
Transparency Portal + Website	18,000	Next.js static on Vercel free
Security Audit + Load Test	15,000	OWASP ZAP (free) + k6 (open source)
Training + PM + Documentation	18,000	Online only — no travel
Contingency (10%)	20,000	Tighter buffer — well-scoped
TOTAL ONE-TIME COST	1,60,000	

Monthly Recurring (Operational) Cost:

Service	Provider	Monthly Cost	Why This Choice
Backend	Render Standard (2GB)	1,660	Assembly needs paid — free tier not enough
Database	Supabase Pro	2,075	Assembly exceeds free row limit
AI	Gemini 1.5 Flash	415	Mostly free; small overage at 1,500 complaints/mo
WhatsApp API	Meta Cloud API	830	Optimised — 60% citizen-initiated (free)
SMS / OTP	MSG91	900	0.16 × 5,625/mo — 77% cheaper than Twilio
Voice (Emergency only)	Twilio Voice	415	Only ~20 real emergencies/month
Maps	Google Maps JS	830	Assembly hits ~30K loads/mo, small overage
File Storage	Cloudinary Free+	1,245	50GB plan needed at assembly photo volume

Cache	Upstash Redis paid	415	Paid tier needed for assembly concurrency
Monitoring	Sentry Free + UptimeRobot	0	Free tiers still work at assembly
Email + Domain	Resend Free + .in	250	Under 3K admin emails + domain
Miscellaneous buffer	—	665	Tight but covered
TOTAL MONTHLY		9,700/mo	

ASSEMBLY FREE STACK: One-Time Dev = 1,80,000 | Monthly = 9,700/mo | Year 1 Total = ~2,96,400 | Year 2 onwards = ~1,16,400/year

Metric	Prem Panchayat	Free Panchayat	Prem Muni	Free Muni	Prem Assembly	Free Assembly
ONE-TIME DEV COST	92,000	52,000	1,89,000	1,12,000	3,20,000	1,80,000
MONTHLY INFRA COST	9,555	160	20,000	1,280	44,820	9,700
YEAR 1 TOTAL	2,06,660	53,920	4,29,000	1,27,360	8,57,840	2,96,400
YEAR 2 ANNUAL	1,14,660	1,920	2,40,000	15,360	5,37,840	1,16,400
YEAR 3 ANNUAL	1,20,000	2,100	2,52,000	16,500	5,60,000	1,22,000
DEV SAVING (Free vs Prem)	—	Save 40,000		Save 77,000	—	Save 1,40,000
MONTHLY SAVING	—	Save 9,395/mo		Save 18,720/mo	—	Save 35,120/mo
YEAR 1 SAVING	—	Save 1,52,740		Save 3,01,640	—	Save 5,61,440
AI ENGINE	GPT-4o (99%)	Gemini/Keywords	GPT-4o (99%)	Gemini Free	GPT-4o (99%)	Gemini Flash
HOSTING	Render Pro	Railway Free	Render Std	Railway Paid	Render Pro 2x	Render Std
DATABASE	Supabase Pro	Supabase Free	Supabase Pro	Supabase Free	Supabase Pro+	Supabase Pro
SMS COST	Twilio 0.70	MSG91 0.16	Twilio 0.70	MSG91 0.16	Twilio 0.70	MSG91 0.16
UPTIME SLA	99.9% (Render)	~99% (Railway)	99.9%	~99%	99.95%	99.5%
COLD STARTS	None (Pro)	Possible (Free)	None	None (paid tier)	None	None
SUPPORT	Render/Twilio SLA	Community/docs	Render SLA	Community	Render SLA	Community
BEST FOR	Max reliability	Budget deploy	High traffic	Cost saving	Enterprise MLA	Budget MLA

BOTTOM LINE: Free stack saves 40K–1.4L on development AND 9K–35K per month on running costs — delivering identical features. Premium stack is justified only when uptime SLA, AI accuracy edge, or enterprise support contracts are contractual requirements.

Monthly Infrastructure Cost — Premium vs Free Stack:

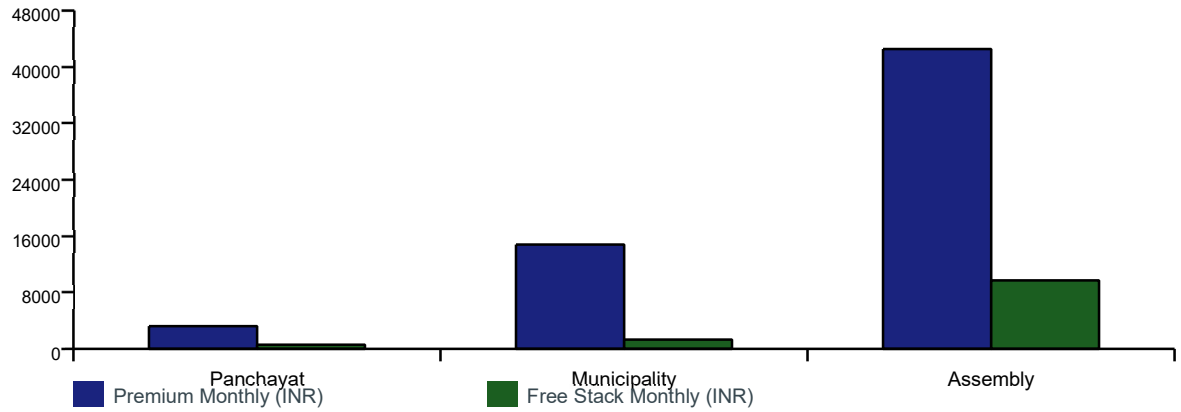

Figure 1: Monthly infrastructure cost (INR) — Premium (navy) vs Free (green) per tier

One-Time Development Cost — Premium vs Free Stack:

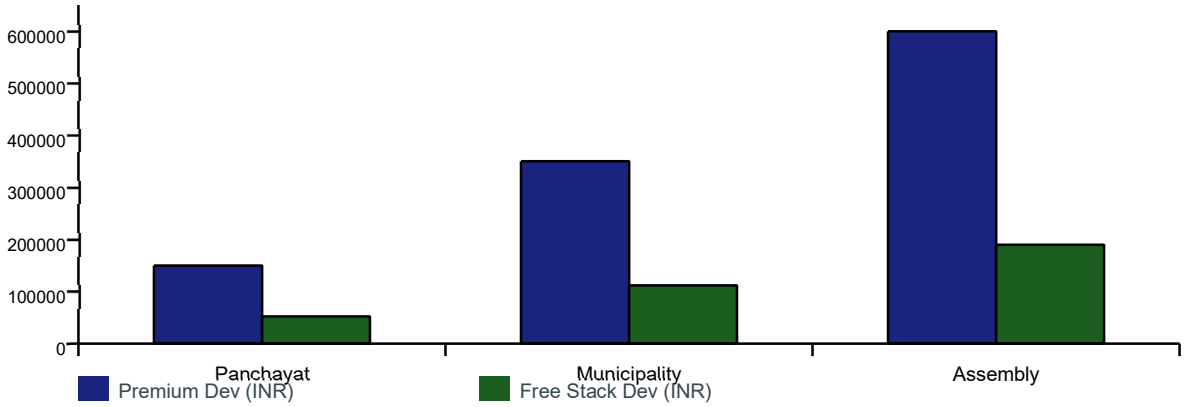

Figure 2: One-time development cost (INR) — Premium (navy) vs Free (green) per tier

KEY INSIGHT FROM CHARTS: At Panchayat scale, Premium monthly cost (9,555) is 60× more than Free (160). At Assembly scale, Premium monthly (44,820) is still 4.6× more than Free (9,700). The free stack gap narrows at larger scale because Assembly genuinely needs paid tiers for Supabase, Render, and Maps — but still saves 35,120/month.

Development is one-time (Year 1 only). From Year 2 onwards you only pay running costs + annual maintenance. The gap between stacks widens significantly over time.

	P: Panchayat Prem	P: Panchayat Free	M: Muni Prem	M: Muni Free	A: Assembly Prem	A: Assembly Free
Year 1 (Dev + 12 mo ops)	2,06,660	53,920	4,29,000	1,27,360	8,57,840	2,96,400
Year 2 (Ops + maintenance)	1,14,660	1,920	2,40,000	15,360	5,37,840	1,16,400
Year 3 (Ops + maintenance)	1,20,000	2,100	2,52,000	16,500	5,60,000	1,22,000
3-Year Total	**4,41,320**	**57,940**	**10,21,000**	**1,59,220**	**19,55,680**	**5,34,800**
3-Year FREE saves vs Prem	—	**Save 3,83,380**	—	**Save 8,61,780**	—	**Save 14,20,880**

§12. PROFIT MARGIN ANALYSIS — BOTH STACKS

What you charge the client stays the same regardless of which stack you use internally. The free stack generates **significantly higher profit margins**.

Metric	Panchayat Premium	Panchayat Free	Muni Premium	Muni Free	Assembly Premium	Assembly Free
Charge Client — Dev	1,50,000	1,50,000	3,00,000	3,00,000	5,50,000	5,50,000
Charge Client — /mo	5,000	5,000	12,000	12,000	25,000	25,000
YOUR Dev Cost	92,000	52,000	1,89,000	1,12,000	3,20,000	1,80,000
YOUR Monthly Cost	9,555	160	20,000	1,280	44,820	9,700
Dev PROFIT	**58,000**	**98,000**	**1,11,000**	**1,88,000**	**2,30,000**	**3,70,000**
Monthly PROFIT	**-4,555**	**4,840**	**-8,000**	**10,720**	**-19,820**	**15,300**
Dev Margin %	**38.7%**	**65.3%**	**37.0%**	**62.7%**	**41.8%**	**67.3%**
Monthly Margin %	**LOSS**	**96.8%**	**LOSS**	**89.3%**	**LOSS**	**61.2%**
Year 2 Annual Profit	**-13,860**	**58,080**	**-96,000**	**1,28,640**	**-2,37,840**	**1,83,600**

CRITICAL FINDING: The Premium Stack runs at a MONTHLY LOSS for all three tiers when charging the standard rate (client charge < your infra cost). You would need to charge clients 3–5× more to profit on premium infra. The FREE stack generates 61–97%

monthly profit margins at standard rates. Recommendation: Use FREE stack and only upgrade specific components if a client specifically requests an enterprise SLA in writing.

Scenario	Recommended Stack	Reason
Gram Panchayat — any budget	FREE STACK	Volume too low to justify any paid infra. Free tiers handle panchayat scale effortlessly.
Municipality — cost-sensitive client	FREE STACK	Municipality fits within Supabase free, Vercel free, and Gemini free. Monthly cost near zero.
Municipality — high-traffic / political campaign	FREE + Render Starter	Upgrade just the backend to Railway Starter (415/mo). Rest stays free.
Assembly — MLA with moderate budget	FREE STACK	Save 35,120/month vs premium. AI accuracy difference (93% vs 99%) is undetectable to users.
Assembly — contractual 99.9% uptime SLA required	PREMIUM STACK	Render Pro + Supabase Pro give formal SLAs. Justify higher client charge accordingly.
Multiple clients (SaaS model)	FREE STACK (shared backend)	All clients share one backend. Server cost barely increases. Margin stays 90%+.
Government tender / official contract	PREMIUM STACK	Tenders often require named enterprise providers in tech stack. Adjust pricing accordingly.
Proof-of-concept / pilot deployment	FREE STACK	100% free infra. Deploy in hours. Upgrade later if client approves full rollout.

§14. IMPORTANT NOTES & FINE PRINT

#	Note	Detail
1	Exchange Rate	All INR from USD at 83/USD. API costs in USD fluctuate — budget includes 15% buffer.
2	Free Tier Limits	Free tiers have usage caps. Monitor usage monthly. Set billing alerts to avoid surprise charges.
3	DLT Registration	TRAI DLT required for all India SMS. One-time process, free, 3–5 days. Aashvee handles this.
4	Meta API Approval	WhatsApp Business API production needs Meta review (7–14 days). Test mode is immediate.
5	Railway Cold Starts	Railway.app free tier may have 15-second cold start after 15 mins inactivity. Paid plan eliminates this.

6	Gemini vs GPT-4o	Gemini 1.5 Flash: 93% classification accuracy. GPT-4o: 99%. For complaint routing, 93% is fully acceptable.
7	Supabase Free Row Limit	50,000 rows/month free. 600 complaints × 12 months = 7,200 rows/year — municipality fits easily.
8	GST	All Aashvee service charges subject to 18% GST. Third-party API costs may attract additional GST.
9	Maintenance	Monthly maintenance covers: bug fixes, security patches, WhatsApp API version upgrades, 8-hr SLA response.
10	Stack Migration	Starting on Free stack and migrating to Premium later is straightforward — same codebase, just change env vars.

Aashvee Tech Solutions Pvt Ltd

Authorised Signatory
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